



Make A Crystal-Locked FM Transmitter

Joy Mukherji

The circuit presented here uses a crystal oscillator and frequency multiplier to generate a highly-stable carrier signal frequency of 96MHz. With 40mW of RF output, it can be used to transmit voice or music up to hundred metres.

The circuit diagram of the 96MHz crystal-locked FM transmitter is shown in Fig. 1. The circuit is built around a low-power audio ampli-

fier using LM386 (IC1), transistors 2N3904 (T1 and T2) and 2N3866 (T3), 24MHz crystals (XTAL1 and XTAL2), rectifier diode 1N4007 (D1), signal diode 1N4148 (D2) and a few other components.

A low-power 24MHz Colpitts oscillator is built around T1 along with biasing resistors R4, R5 and R6, and capacitors C6, C7 and C8.

Transistor T2 is a buffer amplifier

and its collector is rich in harmonics. The tank circuit comprising inductor L1, capacitors C9 and C10, and trimmer capacitor VC1 is tuned to four times the oscillator frequency, that is, 96MHz. T2 boosts the fourth harmonic of 96MHz and suppresses the 24MHz fundamental, second (48MHz) and third (72MHz) harmonics by at least -24dBc, or 250 times.

Inductor L2 and capacitor C13

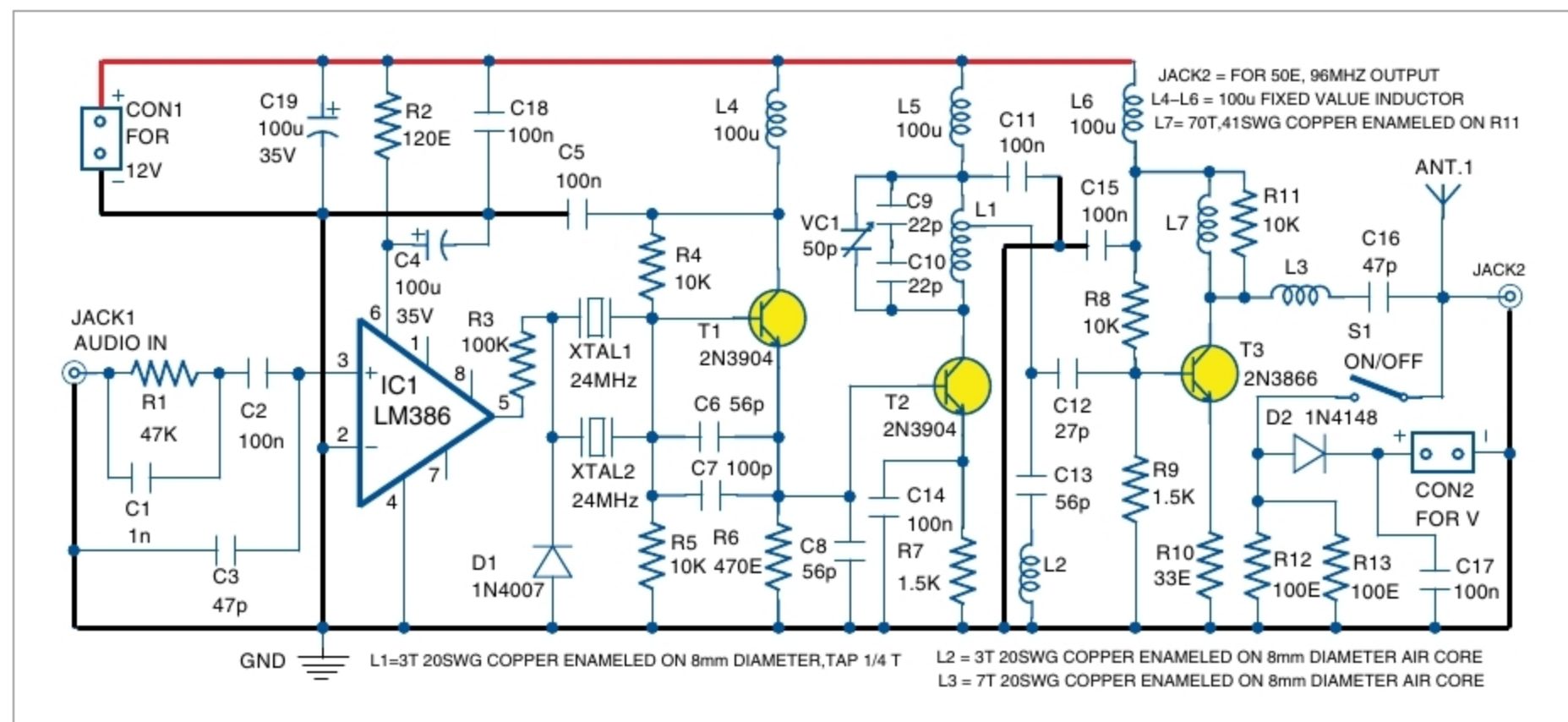


Fig. 1: Circuit diagram of the FM transmitter

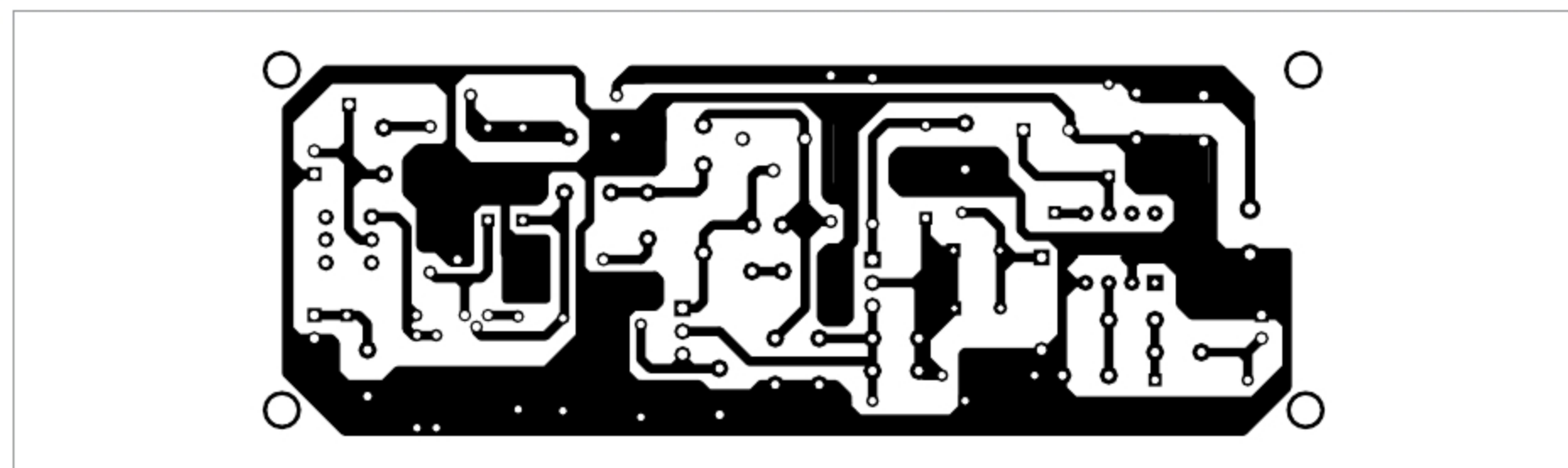


Fig. 2: Actual-size PCB layout of the FM transmitter